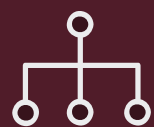


# MODULE INTEREST EARNS RETURNS

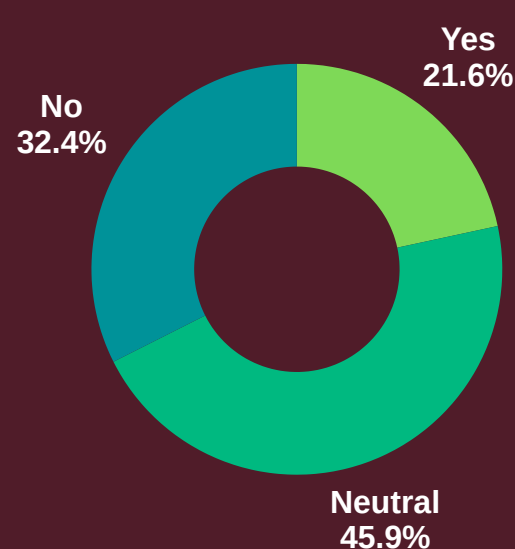
## 1 BACKGROUND



The number of students enrolled in Operations Research 244 has grown significantly over the past 5 years. However, the retention of the number of students enrolling for third-year modules the year after has seen a dip (see bar chart below).



The majority of the class does not major in Operations Research (OR), with only 5% of the current cohort explicitly majoring in OR. From a poll taken in class (using the *Mentimeter* platform), the question was posed if the students have an interest in this module or interests elsewhere, which prompted the following results:



## 2 PROBLEM DESCRIPTION

The results from these statistics formed the basis for the learning outcome. This study focused on improving student engagement and interest in the module by providing students with an opportunity to interact with literature and real-world problems within the field of OR. The learning opportunity allows students to take responsibility for their own learning, and within an area that sparks their individual interests. The overall objectives are to:

1. Promote student involvement in the learning process,
2. Expose students to the graduate attributes (listed below) in a classroom environment,
3. Apply concepts of the class material to real-world scenarios,
4. Cultivate interest for the field of OR in academia.

## 3 METHODOLOGY



The learning outcome is presented as a research assignment. The aim of the assignment is to research an area (of the student's choosing) where they can apply techniques for optimisation using the knowledge they have acquired from the module or any other OR module.

- The assignment must be submitted in the form of a research proposal.
- Marks will be awarded for thorough researching of the area of study, identifying a problem where OR can be useful.
- Students must identify objectives, variables and potential constraints that may be encountered.

The assignment was broadly stated to allow the students to identify a topic that interests them. However, the necessary criteria from the research proposal were clearly stated.

According to Brew (2013), the concept of undergraduate research "emphasise[s] student engagement, participation and inquiry". It presents students with the opportunity to critically engage with the class content and apply it to a real world context.

Students were also asked to participate in a word cloud to comment on their experience of the module, and any phrases that can help the lecturer identify improvements or concerns from students' perspectives.

## 4 FINDINGS



The quality of the research proposals were of a high standard, which suggests that students took the time with the assignment. There was evidence of consulting literature, and every student identified a unique topic which sparked their interest.

Besides that the assignment forms part of formative assessments, it has also allowed students to be reflective of the content they have been exposed to within their academic careers

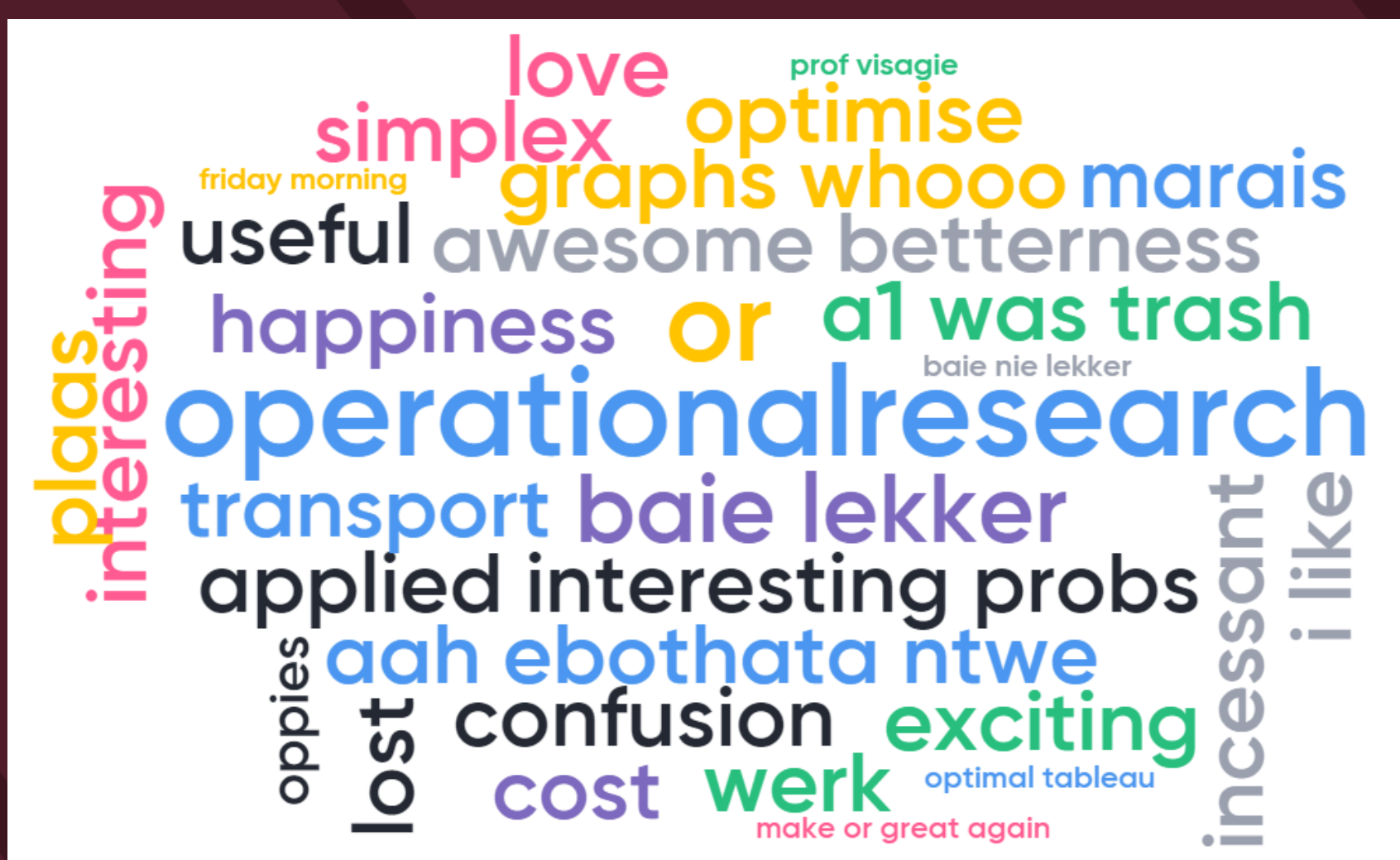
The *Mentimeter* platform is a useful tool for live interaction between students, the lecturer and the content. It does, however, bear the potential to be misused if the questions are not clearly stated.

## 5 CONCLUSION



A learning opportunity of this nature can assist lecturers to identify if students understand the content. The length of the proposal should be stated clearly.

The results on whether the retention of students onto third-year level will only be confirmed in the next year. Even if the retention is not improved, an assignment such as this promotes an enquiring mind, engaged citizenship, innovation and problem-solving skills through inquiry-based and research-based learning.



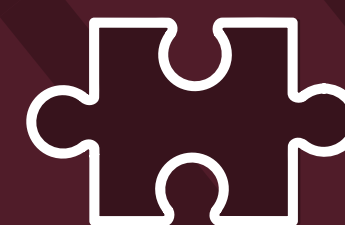
AN ENQUIRING  
MIND



AN ENGAGED  
CITIZEN



A DYNAMIC  
PROFESSIONAL



A WELL-ROUNDED  
INDIVIDUAL



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